

Pre- Feasibility Study Seabream Fish Farm



2017

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1. Executive Summary

The objective of the project is to construct Seabream fish farm in Aqaba city, which is located in the south of the hashemite kingdom of Jordan. Aqaba is famous for its views of the Red Sea. Imported fishes got the highest market share of the local market, hence araised the idea of establishing a fish breeding farm to cover the high demad of fishes in Jordan and, to achive self sufficiency of Seabream fishes. Moreover, this project will provide 18 job opportunities, leading to enhance the economic situation in the region.

The following tables shows the results of the study:

Table1 main results of the project

Results	
Total investment cost (JD)	1,478,115
IRR (%)	%23.13
Payback period (years)	5
Net present value (for 10 years)	809,458
Provided job opportunities	18
Expected total cash inflows (for 10 years)	13,600,876
Total operating expenses (for 10 years)	9,142,384
Total net profit (for 10 years)	2,297,085

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

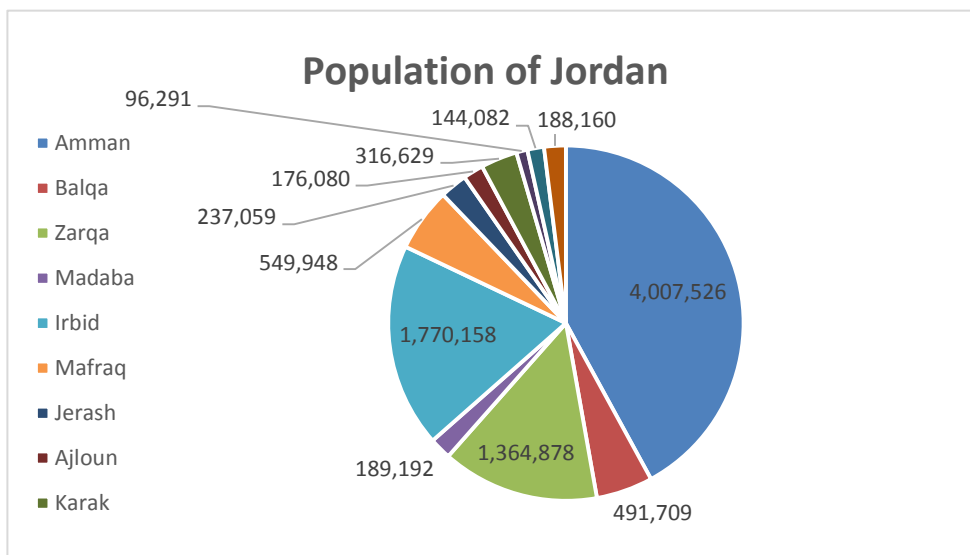
Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the

Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

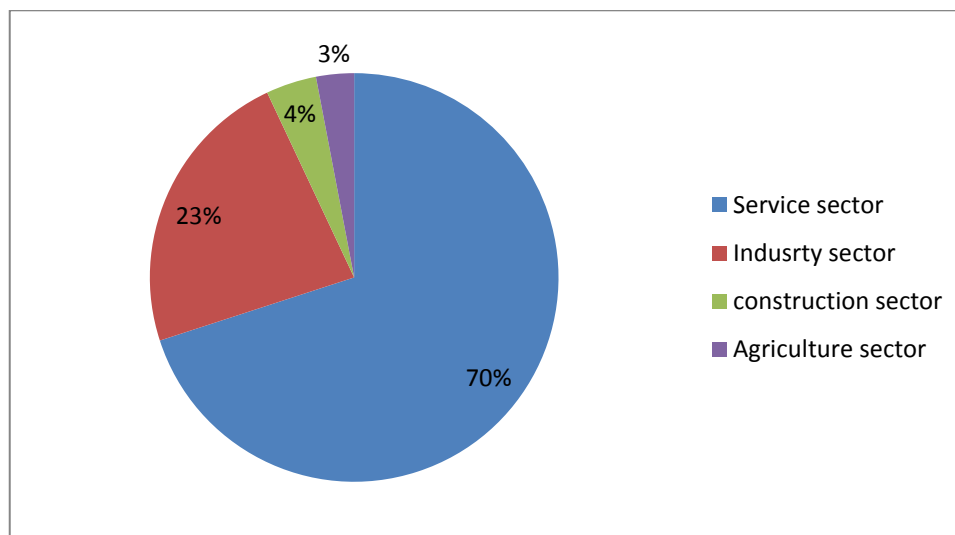
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally the new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non-residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions. Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.

- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The idea of this project is to breed Seabream fishes in an intensive way in Aqaba Governorate. The project must consist of several basins that are made of fiber glass which will be placed in plastic houses to maintain a certain temperature during the year. Water is injected with liquid oxygen to raise the proportion of dissolved oxygen in the water. Every basin will be provided with an automatic feeding system to feed the fishes during the whole day, All the operations and functions will be controlled by the central computer.

Seabream fishes are bisexual ceatures, at the beginning of their life they are males (during the first or the second), and then they transform into females in the second or third year, and the transformation of sex are subject to environmental and social conditions. The eggs are placed usually from October to December, nursery continues for two days at a temperature of 16-17 degrees Celsius. And the period of larvae stays about 50 days at 17.5 degrees Celsius, or about 43 days at 20 degrees Celsius.

It is proposed to establish 12 basins on the selected land, and the project contains 4 plastic house each house contains 3 basins. The eggs should stay under standard conditions until they are grown to a fish and that takes almost two months, then they are moved to the baisns that contains the fishes that are ready to be sold. The project requires pumps to transfer the water of the Sea to the baisns, because the sea water is essential for Seabream fishes to live.

Objectives

The objective of the project can be sumerized in the following points:

- Covering the high demand of fishes in Jordan.
- Trying to achive Self-sufficiency from Seabream fishes, and exporting to other countries.
- Providing jobs oppourtunities, and improving the economic situation in Jordan.
- Achieving good profits from this project.

Proposed Product

The project provides Seabream fishes to the local market and trying to export to other countries. As it is noted that there is high demand of Seabream fishes in Jordan, and almost all of Seabream fishes are imported from outside.

Thus, arises the need of such project to provide this product of Seabream fishes, to cover the local demand and to achieve Self-sufficiency from Seabream fish.

Target Segments

The project targets the following segments:

- The local market.
- External markets (export to the other countries) in the coming years.

Market Size Analysis

The market size has been studied depending on the amount of imported and exported fishes in the local market in 2015, the following table shows the amount of imported and exported fishes in 2015:

Table 3 Amount of imports and exports in 2015

product	Imported (JD)	Imported (KG)	exported (JD)	exported (KG)	Re- exported (JD)	Re- exported (KG)	Total exported (JD)	Trade balance (JD)
Fresh fish or chilled, excluding fish fillet and other fish meat of heading	12,502,536	3,000,244	0	0	74,148	31,000	74,148	-12,428,388
Frozen fish except slides and other fish meat of heading	17,270,516	12,315,231	0	0	1,610,996	1,059,220	1,610,996	-15,659,520
Fillets and other fish meat (even if minced), fresh, chilled or frozen	7,341,079	3,628,149	0	0	218,733	146,250	218,733	-7,122,346
Dried fish, salted or in brine, smoked fish, whether or not cooked before or during the smoking process	1,016,028	258,942	0	0	0	0	0	-1,016,028

Department of statistics/Economic sector 2015

After studying and analyzing the local market, Jordan's exports of live fish were almost nonexistent. The Jordanian market is covered by fishing boats and fish farms. Imported fishes got the highest share of the local market. Imported fish cover the largest share of the local market demand and some of the imported fish are re-exported to neighboring countries. Department of Statistics released a report on the preliminary results of fishes survey (fishery farms, and fishing) for 2015. The report pointed out that the amount of fish produced amounted to 1,154 tonnes in 2015, with an increase of 0.8% compared with 2014. The results showed that the amount of fish produced of fish farms breeding has increased by 5.6% during 2015 compared with 2014, reaching quantity 949.25 tonnes in 2015 compared with 898.8 tonnes in 2014. The report pointed out that the fish that have been bred in fish farms breeding has been narrowed to two Fish only two carp and comb, which consisted a percent of 66% and 34% respectively of the total by 2015. Where the first scan covers all the fish breeding farms and of the 34 farms which are visited once every six months, while the second scan covers all fishing boats in Aqaba which are 62 boats, that are being visited monthly.

Competitors analysis

At the moment, there are 34 fishery farms in Jordan containing different kinds of fishes nevertheless, taking into consideration that there aren't any Seabream fisheries in Jordan, and that imports are the only source of seabream. Fish farms are considered the direct competitors for the suggested farm. Fish dealers and fishing boats are considered to be undirect competitors, by importing fresh and frozen fishes to the local market.

Capacity

It is expected farm capacity will be 400 tones of seabream fishes, the capacity is assumed based on production of similar farms.

Prices analysis and pricing policy

After studying the local fish market, the average selling price is 4.5-5 JD per kilo for seabream fish. For that we recommended that the selling price is 3.25 JD per kilo, and this price considered to be competitive, for the product is new and to enable it to compete in the local market.

Income Analysis

The following table shows the annual revenues for 10 year:

Table 4 expected revenues for 10 years (2018-2027)

year	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected Income	1,300,000	1,313,000	1,326,130	1,339,391	1,352,785	1,366,313	1,379,976	1,393,776	1,407,713	1,421,790

The annual revenues were calculated depending on the expected selling capacity and price, and that is shown in the next table:

Table 5 production capacity analysis

Production capacity analysis	
Annual selling quantity	400000
Price per kilo	3.25

4. Technical Analysis

Required Human Resources

Table 6 required human resources

Job position	Number
General manager	1
Agricultural engineer	1
Accountant	1
Marketing officer	1
secretary	1
Warehouse	1
Technical support	1
Total indirect employees	7
Animal production	1
workers	6
Maintinace employee	1
driver	2
cleaners	1
Total direct employees	11
Total employees	18

The following is the job description for the needed employees:

Location Analysis

Aqaba is considered a strategic location to build up the project because of it's closeness to the Red Sea, which facilitates the process of the withdrawal of the Red Sea water to fill the basins, and because of the easiness of exportation to other countries in future.

Description of The Production Process

Taking care of the eggs until they become larvae, under standard conditions such as specific temperature and a high percentage of oxygen. So, it's essential to have automatic feeding machines and Liquid oxygen pump. After two months of high care, fishes become mature and ready to be transported to the basins that contains fishes that are ready to be sold. Oxygen percentage and temperature are checked by sensors devices. It's important to replace the basins' water every 3 days.

Technical Requirements Analysis

1.1.1 Land and construction works

The project will require a land with an area of 20,000 m² Outside the the organizational chart of Aqaba Governorate, the following table shows the deatails of the required land:

Table 7 required land

The land	Area (m ²)
Required land	20,000

The construction works that are needed is shown in the following table:

Table 8 required construction work

Construction works	Area(m ²)
Construction Works	1,500
Design and blueprints	1,500
Other Works	-

Machinery and equipment

The following table shows the required machinery and equipment for the project:

Table 9 required machinery and equipment

Machinery and equipment	Qty
Computers	4
Air conditions	2
Plastic houses	6
Fiber glass baths	12
Automatic feeding machine	12
Measurments and sensors devices	12
Liquid oxygen tank + pump + pipes	12
Central computer and control system	1

Furniture

The required furniture is shown in the following table:

Table 10 required furniture

Required Furniture	Qty
Furniture	-

Vechiles and Transportation

The following table shows the required vechiles:

Table 11 required vehicles for the project

Vechiles	Qty
Pickup	2

5. Financial analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 12 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14 working time assumption

Work time	
Number of weeks in the year	52
Number of working days in the week	7
Number of days in the year	360
Daily working hours	8

Table 15 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	1.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 16 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	50%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	0.5%
Depreciation Expenses from Total Revenues	0.5%
Insurance Expenses from Total Assets Value	0.5%
Marketing Expenses from Total Revenues	1.0%

Table 17 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	0%
Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 18 Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 19 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	591,246
Loans Value	886,869
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	886,869
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	591,246
Equity Percentage	%40
Gross	
Project Value	1,478,115
Weighted Average Cost of Capital	%12.05

Capital Expenditures

The estimated cost of the project is 1,478,115 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 20 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	20	20,000	400,000
Total Cost			400,000

Table 21 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	/ Gross Cost Jordan Dinar
Construction Works	250	1,500	375,000
Design and blueprints	7	1,500	10,500
Other Works	0	-	37,500
Total			423,000

Table 22 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Computers	4	350	1,400
Air conditions	2	500	1,000
Plastic houses	6	4,000	24,000
Fiber glass baths	12	10,000	120,000
Automatic feeding machine	12	5,000	60,000
Measurments and sensors devices	12	1,000	12,000
Liquid oxygen tank + pump + pipes	12	200	2,400
Central computer and control system	1	3,000	3,000
Total			223,800

Table 23 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
General furniture	-	-	6,000
Total			6,000

Table 24 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Medium vehicle	2	23,400	46,800
Total cost			46,800

Table 25 Pre-Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	1,000
Transportation	250
Marketing	500
Training	500
Total cost	2,250

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 26 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	216,040
General and administrative expenses	3	%25	4,001
Marketing expenses	3	%25	3,250
Indirect Salaries	3	%25	18,600
Total			241,891

Table 27 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	400,000	10%	440,000	%29.8
Construction works	423,000	10%	465,300	%31.5
Machinery & Equipment	223,800	10%	246,180	%16.7
Furniture	6,000	10%	6,600	%0.4
Vehicles	46,800	10%	51,480	%3.5
Pre-Operating Expenses	2,250	10%	2,475	%0.2
Working capital	241,891	10%	266,080	%18.0
Total (JD)	1,343,741		1,478,115	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,478,115) JD, where 60% of the project will be financed through loans (886,869) JD, and the remaining (40%) through (Equity) with (591,246) JD. The following table summarizes the general structure of the required financing:

Table 28 Sources of Funding

Funding Sources	Amount	Percentage
Equity	591,246	%40
Loans	886,869	%60
Total	1,478,115	%100
Use of Fund	Amount	Percentage
Capital Expenditures	1,209,560	%81.83
Pre-Operating Expenses	2,475	%0.17
Working Capital	266,080	%18.00
Total	1,478,115	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 29 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	650,000	656,500	663,065	669,696	676,393	683,157	689,988	696,888	703,857	710,895
Service benefit expenses from revenues	65,000	65,650	66,307	66,970	67,639	68,316	68,999	69,689	70,386	71,090
Maintenance from Revenues	6,500	6,565	6,631	6,697	6,764	6,832	6,900	6,969	7,039	7,109
Disposables from Revenues	6,500	6,565	6,631	6,697	6,764	6,832	6,900	6,969	7,039	7,109
Direct Salaries	57,600	59,962	62,420	64,979	67,643	70,417	73,304	76,309	79,438	82,695
Others	78,560	79,524	80,505	81,504	82,520	83,555	84,609	85,682	86,776	87,890
Total	864,160	874,766	885,558	896,542	907,723	919,107	930,700	942,506	954,533	966,788

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 30 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,400	2,448	2,497	2,547	2,598	2,650	2,703	2,757	2,812	2,868
Hospitality	1,200	1,224	1,248	1,273	1,299	1,325	1,351	1,378	1,406	1,434
Stationary and Printing	600	612	624	637	649	662	676	689	703	717
Transportation	3,800	3,876	3,954	4,033	4,113	4,196	4,279	4,365	4,452	4,541
Training	500	510	520	531	541	552	563	574	586	598
Insurance	6,048	6,169	6,292	6,418	6,546	6,677	6,811	6,947	7,086	7,228
Miscellaneous	1,455	1,484	1,514	1,544	1,575	1,606	1,638	1,671	1,705	1,739
Total	16,003	16,323	16,649	16,982	17,322	17,668	18,022	18,382	18,750	19,125

Marketing Expenses

Table 31 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	13,000	13,130	13,261	13,394	13,528	13,663	13,800	13,938	14,077	14,218

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance, etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 32 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
General manager	1	1	1	1	1	1	1	1	1	1
Agricultural engineer	1	1	1	1	1	1	1	1	1	1
Accountant	1	1	1	1	1	1	1	1	1	1
Marketing officer	1	1	1	1	1	1	1	1	1	1
secretary	1	1	1	1	1	1	1	1	1	1
Warehouse	1	1	1	1	1	1	1	1	1	1
Technical support	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	7	7	7	7	7	7	7	7	7	7
direct employees										
Animal production	1	1	1	1	1	1	1	1	1	1
workers	6	6	6	6	6	6	6	6	6	6
Maintenance technician	1	1	1	1	1	1	1	1	1	1
driver	2	2	2	2	2	2	2	2	2	2

cleaners	1	1	1	1	1	1	1	1	1	1
Total number of direct employees	11	11	11	11	11	11	11	11	11	11
Grand total	18	18	18	18	18	18	18	18	18	18

Table 33 Expected monthly Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
General manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Agricultural engineer	750	781	813	846	881	917	954	994	1,034	1,077
Accountant	500	521	542	564	587	611	636	662	690	718
Marketing officer	500	521	542	564	587	611	636	662	690	718
secretary	400	416	433	451	470	489	509	530	552	574
Warehouse	500	521	542	564	587	611	636	662	690	718
Technical support	500	521	542	564	587	611	636	662	690	718
direct employees										
Animal production	500	521	542	564	587	611	636	662	690	718
workers	300	312	325	338	352	367	382	397	414	431
Maintenance technician	350	364	379	395	411	428	445	464	483	502
driver	350	364	379	395	411	428	445	464	483	502
cleaners	250	260	271	282	294	306	318	331	345	359

Table 34 Expected Annual Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
General manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Agricultural engineer	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Accountant	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Marketing officer	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
secretary	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Warehouse	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Technical support	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Total indirect salaries	74,400	77,450	80,626	83,932	87,373	90,955	94,684	98,566	102,607	106,814
direct employees										
Animal production	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
workers	28,800	29,981	31,210	32,490	33,822	35,208	36,652	38,155	39,719	41,347
Maintenance technician	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
driver	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
cleaners	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Total direct salaries	57,600	59,962	62,420	64,979	67,643	70,417	73,304	76,309	79,438	82,695
Grand total	132,000	137,412	143,046	148,911	155,016	161,372	167,988	174,876	182,045	189,509

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 35 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	440,000	%0.0	%0.0
Construction Works	465,300	%5.0	%0.0
Machinaries	246,180	%10.0	%2.0
Furniture	6,600	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0
	1,209,560		

Table 36 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,209,560	1,214,814	1,220,182	1,225,669	1,231,275	1,237,006	1,242,863	1,248,850	1,254,970	1,261,227
Total Depreciation Expenses	53,691	54,216	54,753	55,302	55,863	56,436	57,021	57,620	58,232	58,858
Total Accumulated Depreciation	53,691	107,907	162,661	217,962	273,825	330,261	387,282	444,902	503,134	561,992
Total Additions	0	5,254	5,369	5,486	5,607	5,731	5,857	5,987	6,120	6,256
Total Net Book Values	1,155,869	1,106,906	1,057,522	1,007,706	957,451	906,745	855,581	803,948	751,836	699,235

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 37 Loan Details

Loan Value		886,869		
Annual Interest Rate		9.48%		
Loans Period		5		
Grace Period		0		
Loan Starts at year:		2018		
Annual Payment		230,854		
Number of Payments		5		
Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				886,869
2019	230,854	84,075	146,779	740,090
2020	230,854	70,161	160,693	579,396
2021	230,854	54,927	175,927	403,469
2022	230,854	38,249	192,605	210,864
2023	230,854	19,990	210,864	0

Income Statement

Table 38 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,300,000	1,313,000	1,326,130	1,339,391	1,352,785	1,366,313	1,379,976	1,393,776	1,407,714	1,421,791
Gross Operating Revenues	1,300,000	1,313,000	1,326,130	1,339,391	1,352,785	1,366,313	1,379,976	1,393,776	1,407,714	1,421,791
Operating Expenses	(864,160)	(874,766)	(885,558)	(896,542)	(907,723)	(919,107)	(930,700)	(942,506)	(954,533)	(966,788)
Gross Operating Profit	435,840	438,234	440,572	442,849	445,062	447,206	449,277	451,270	453,180	455,003
<i>Gross Profit Percentage</i>	<i>%34</i>	<i>%33</i>	<i>%33</i>	<i>%33</i>	<i>%33</i>	<i>%33</i>	<i>%33</i>	<i>%32</i>	<i>%32</i>	<i>%32</i>
Salaries and Benefits (Indirect	(74,400)	(77,450)	(80,626)	(83,932)	(87,373)	(90,955)	(94,684)	(98,566)	(102,607)	(106,814)
General and Administraive	(16,003)	(16,323)	(16,649)	(16,982)	(17,322)	(17,668)	(18,022)	(18,382)	(18,750)	(19,125)
Markeing Expenses	(13,000)	(13,130)	(13,261)	(13,394)	(13,528)	(13,663)	(13,800)	(13,938)	(14,077)	(14,218)
Pre-Operating Expenses	(2,475)									
Gross Indirect Expenses	(105,878)	(106,903)	(110,536)	(114,308)	(118,222)	(122,286)	(126,505)	(130,886)	(135,434)	(140,157)
Income before Interest,	329,962	331,331	330,036	328,542	326,840	324,919	322,771	320,384	317,746	314,846
Fixed Assets Depreciations	(53,691)	(54,216)	(54,753)	(55,302)	(55,863)	(56,436)	(57,021)	(57,620)	(58,232)	(58,858)
Income before Tax and Interests	276,271	277,115	275,282	273,240	270,977	268,484	265,750	262,764	259,514	255,989
Bank Interests	(84,075)	(70,161)	(54,927)	(38,249)	(19,990)	0	0	0	0	0
Income Before Tax	192,196	206,954	220,356	234,991	250,987	268,484	265,750	262,764	259,514	255,989
Income Tax	(9,610)	(10,348)	(11,018)	(11,750)	(12,549)	(13,424)	(13,287)	(13,138)	(12,976)	(12,799)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	182,586	196,607	209,338	223,241	238,438	255,060	252,462	249,626	246,538	243,189
<i>Net Profit Percentage</i>	<i>%14</i>	<i>%15</i>	<i>%16</i>	<i>%17</i>	<i>%18</i>	<i>%19</i>	<i>%18</i>	<i>%18</i>	<i>%18</i>	<i>%17</i>
Compulsory Reserves	(18,259)	(19,661)	(20,934)	(22,324)	(23,844)	(25,506)	(25,246)	(24,963)	(24,654)	(24,319)
Retained Earnings	164,328	341,274	529,678	730,595	945,189	1,174,743	1,401,959	1,626,622	1,848,506	2,067,377

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 39 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	182,586	196,607	209,338	223,241	238,438	255,060	252,462	249,626	246,538	243,189
Bank Interests	84,075	70,161	54,927	38,249	19,990	0	0	0	0	0
Depreciation	53,691	54,216	54,753	55,302	55,863	56,436	57,021	57,620	58,232	58,858
Total Operating Cashflows before Additions to Working Capital	320,353	320,983	319,018	316,792	314,290	311,495	309,484	307,246	304,770	302,047
Inventory (Increase/Decrease)	(54,167)	(542)	(547)	(553)	(558)	(564)	(569)	(575)	(581)	(587)
Accounts Receivables	(108,333)	(1,083)	(1,094)	(1,105)	(1,116)	(1,127)	(1,139)	(1,150)	(1,161)	(1,173)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
(Increase/Decrease)										
Accounts Payables (Increase/Decrease)	53,540	1,026	1,057	1,088	1,121	1,155	1,190	1,227	1,265	1,304
Working Capital (Increase/Decrease)	(108,960)	(599)	(584)	(569)	(553)	(536)	(518)	(498)	(478)	(456)
Net Cashflows from Operating Activities	211,393	320,385	318,433	316,223	313,737	310,959	308,966	306,747	304,293	301,591
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(1,209,560)	(5,254)	(5,369)	(5,486)	(5,607)	(5,731)	(5,857)	(5,987)	(6,120)	(6,256)
Net Cashflows from Investments Activities	(1,209,560)	(5,254)	(5,369)	(5,486)	(5,607)	(5,731)	(5,857)	(5,987)	(6,120)	(6,256)
Cashflows from Financing Activities										
Capital	591,246									
Loan Amortization	(146,779)	(160,693)	(175,927)	(192,605)	(210,864)	0	0	0	0	0
Bank Interest Rate	(84,075)	(70,161)	(54,927)	(38,249)	(19,990)	0	0	0	0	0
Loans	886,869	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,247,261	(230,854)	(230,854)	(230,854)	(230,854)	0	0	0	0	0
Net (Increase/Decrease) in Cash	249,093	84,277	82,211	79,882	77,276	305,229	303,109	300,760	298,173	295,335
Cashflows at the Beginning of Period	0	249,093	333,371	415,581	495,464	572,740	877,969	1,181,077	1,481,838	1,780,010
Cashflows at the End of Period	249,093	333,371	415,581	495,464	572,740	877,969	1,181,077	1,481,838	1,780,010	2,075,345

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 40 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	249,093	333,371	415,581	495,464	572,740	877,969	1,181,077	1,481,838	1,780,010	2,075,345
Inventory	54,167	54,708	55,255	55,808	56,366	56,930	57,499	58,074	58,655	59,241
Accounts Receivable	108,333	109,417	110,511	111,616	112,732	113,859	114,998	116,148	117,309	118,483
Total current Assets	411,593	497,496	581,348	662,888	741,838	1,048,758	1,353,574	1,656,060	1,955,975	2,253,069
Non Current Assets										
Fixed Assets (net)	1,155,869	1,106,906	1,057,522	1,007,706	957,451	906,745	855,581	803,948	751,836	699,235
Total Non Current Assets	1,155,869	1,106,906	1,057,522	1,007,706	957,451	906,745	855,581	803,948	751,836	699,235
Total Assets	1,567,462	1,604,402	1,638,869	1,670,594	1,699,288	1,955,503	2,209,156	2,460,008	2,707,811	2,952,304
Liabilities										
Current Liabilities										
Paybles	53,540	54,566	55,623	56,712	57,833	58,988	60,178	61,405	62,669	63,973
Remaining amount of Loan	160,693	175,927	192,605	210,864	0	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total current Liabilities	214,233	230,494	248,228	267,576	57,833	58,988	60,178	61,405	62,669	63,973
Non Current Liabilities										
Long Terms Loans	579,396	403,469	210,864	0	0	0	0	0	0	0
Total Long Term Liabilities	579,396	403,469	210,864	0	0	0	0	0	0	0
Total Liabilities	793,630	633,963	459,093	267,576	57,833	58,988	60,178	61,405	62,669	63,973
Owners Equity										
Shareholders Contributions	591,246	591,246	591,246	591,246	591,246	591,246	591,246	591,246	591,246	591,246
Statutory Reseve	18,259	37,919	58,853	81,177	105,021	130,527	155,773	180,736	205,390	229,709
Retained Profits	164,328	341,274	529,678	730,595	945,189	1,174,743	1,401,959	1,626,622	1,848,506	2,067,377
Total Equity	773,832	970,439	1,179,777	1,403,018	1,641,456	1,896,515	2,148,978	2,398,603	2,645,142	2,888,331
Total Liabilities and Equity	1,567,462	1,604,402	1,638,869	1,670,594	1,699,288	1,955,503	2,209,156	2,460,008	2,707,811	2,952,304

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans; therefore, the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 41 Free Net Cash Flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(1,478,115)	213,868	315,131	313,065	310,736	308,130	305,229	303,109	300,760	298,173	295,335	3,814,749
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for Cash Flows	(1,478,115)	152,017	199,903	177,233	156,994	138,934	122,823	108,852	96,392	85,284	75,387	973,754

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 42 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(1,478,115)	213,868	315,131	313,065	310,736	308,130	305,229	303,109	300,760	298,173	295,335
Project Value Returned	1,478,115	1,264,247	949,116	636,051	325,314	17,184	(288,044)	(591,153)	(891,913)	(1,190,086)	(1,485,421)
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 43 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	809,458
Payback Period	5
Internal Rate of Return	%23.13

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 44 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%23.13	5	12.05%
Revenues declined by 10%	%19.94	5	12.05%
Operating Expenses Increased by 10%	%16.34	7	12.05%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 809,458 Jordan Dinars considering that the project provides 18 Job Opportunities for the governorate residents.

